

Accident Risk Prediction and Hotspot Analysis



A Machine Learning Based Project Report

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1. Introduction

Road accidents are influenced by multiple dynamic factors such as traffic volume, weather conditions, and temporal patterns. Predicting accident risk in advance can help authorities and road users take preventive measures. This project aims to develop a machine-learning-based system that predicts accident risk levels (Low, Medium, High) and identifies condition-based hotspots. The approach uses PCA for risk scoring, supervised learning models for classification, and KMeans clustering for pattern analysis. The system demonstrates how data-driven approaches can support intelligent transportation and road safety.

2. Literature Survey

Several studies highlight the importance of traffic and weather conditions in accident prediction.

- Machine Learning techniques like Random Forest, Logistic Regression, and SVM have been widely used for traffic-related forecasting.
- PCA is commonly applied for dimensionality reduction and extracting dominant patterns, helping improve prediction stability.
- Clustering techniques (e.g., KMeans) are often used to identify accident-prone zones by grouping similar conditions.
- Recent research focuses on integrating environmental data, temporal factors, and urban mobility datasets to build predictive systems.

These works informed the choice of PCA, supervised ML models, and clustering for this project.

3. Research Gaps & Findings

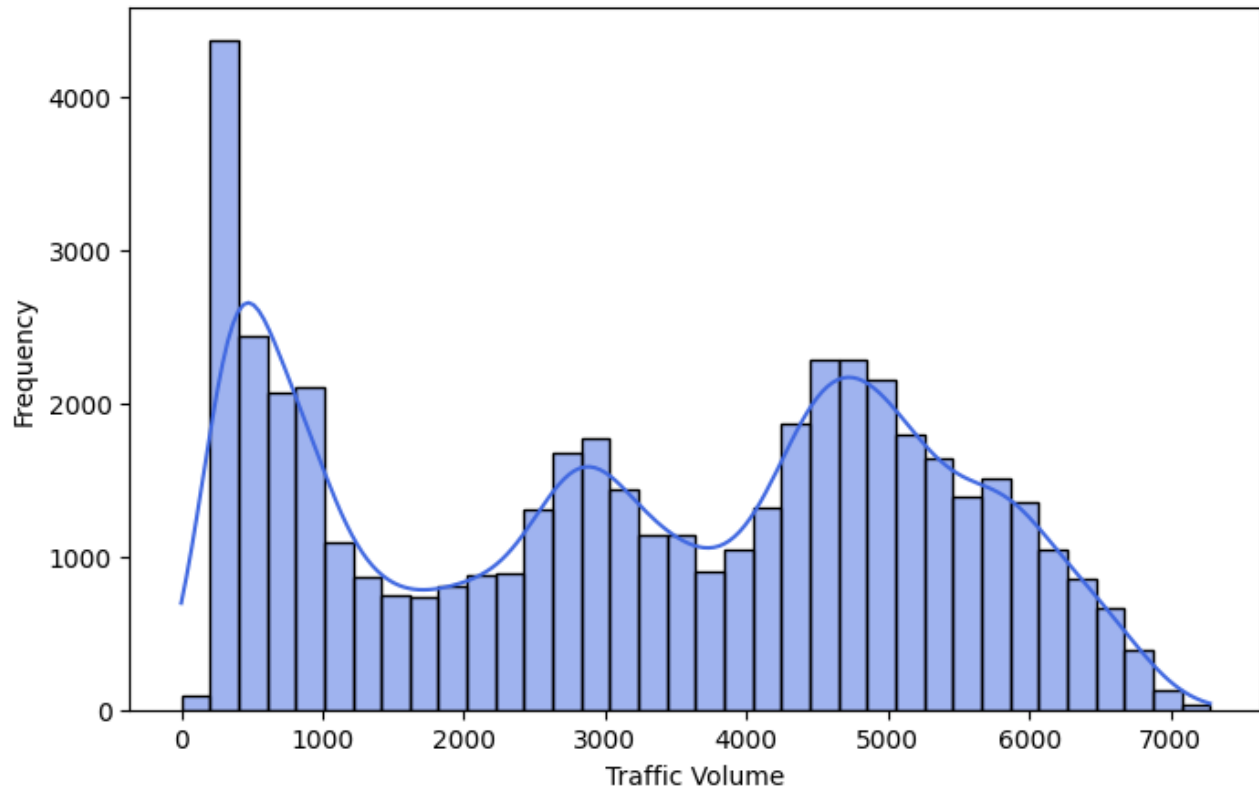
Research Gaps Identified:

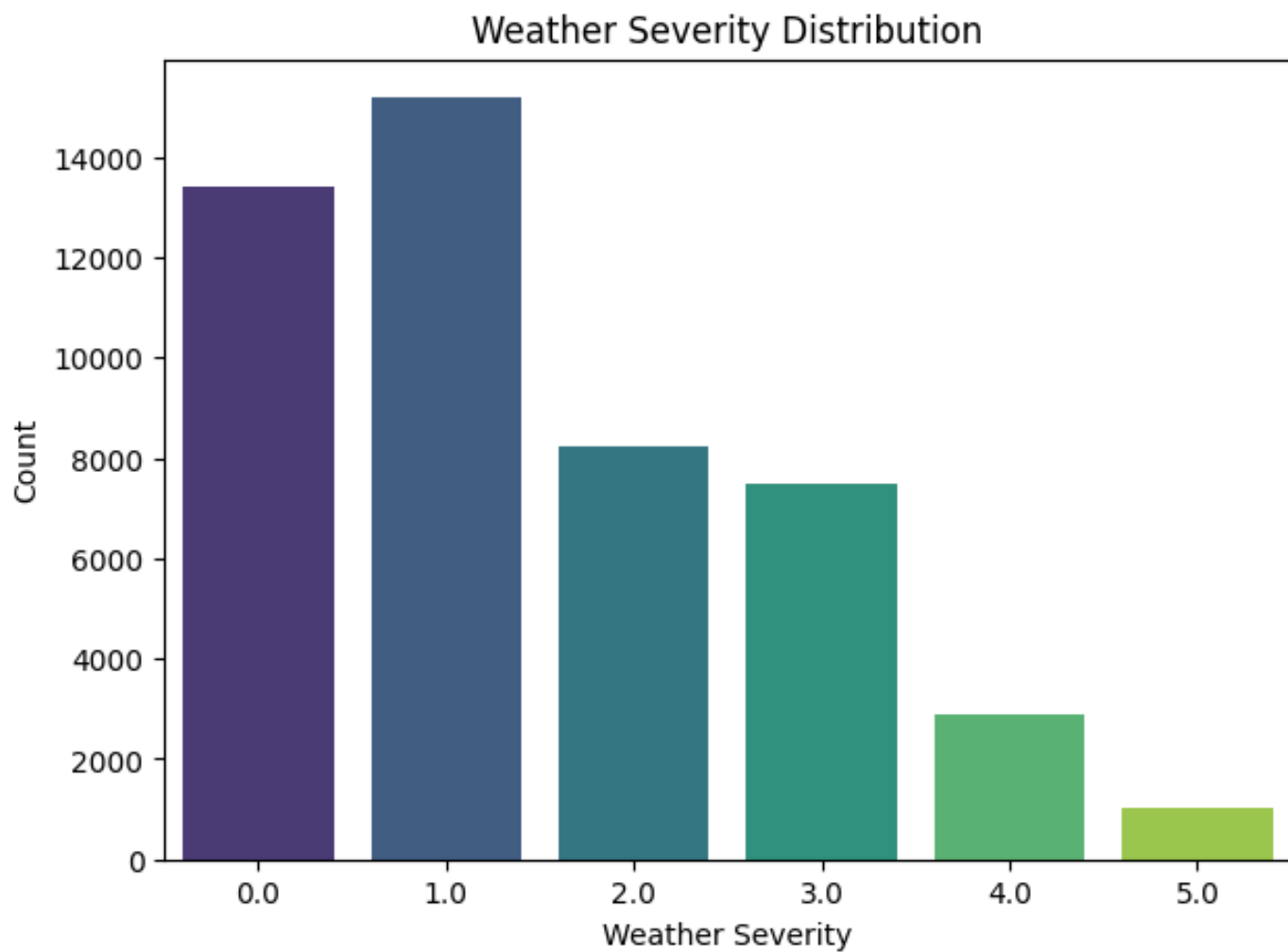
- Most existing studies focus on accident severity after an accident occurs, not *condition-based* pre-accident prediction.
- Manual risk scoring introduces human bias.
- Limited systems combine both prediction and hotspot pattern discovery.
- Few models are designed to be integrated with a real-time user interface.

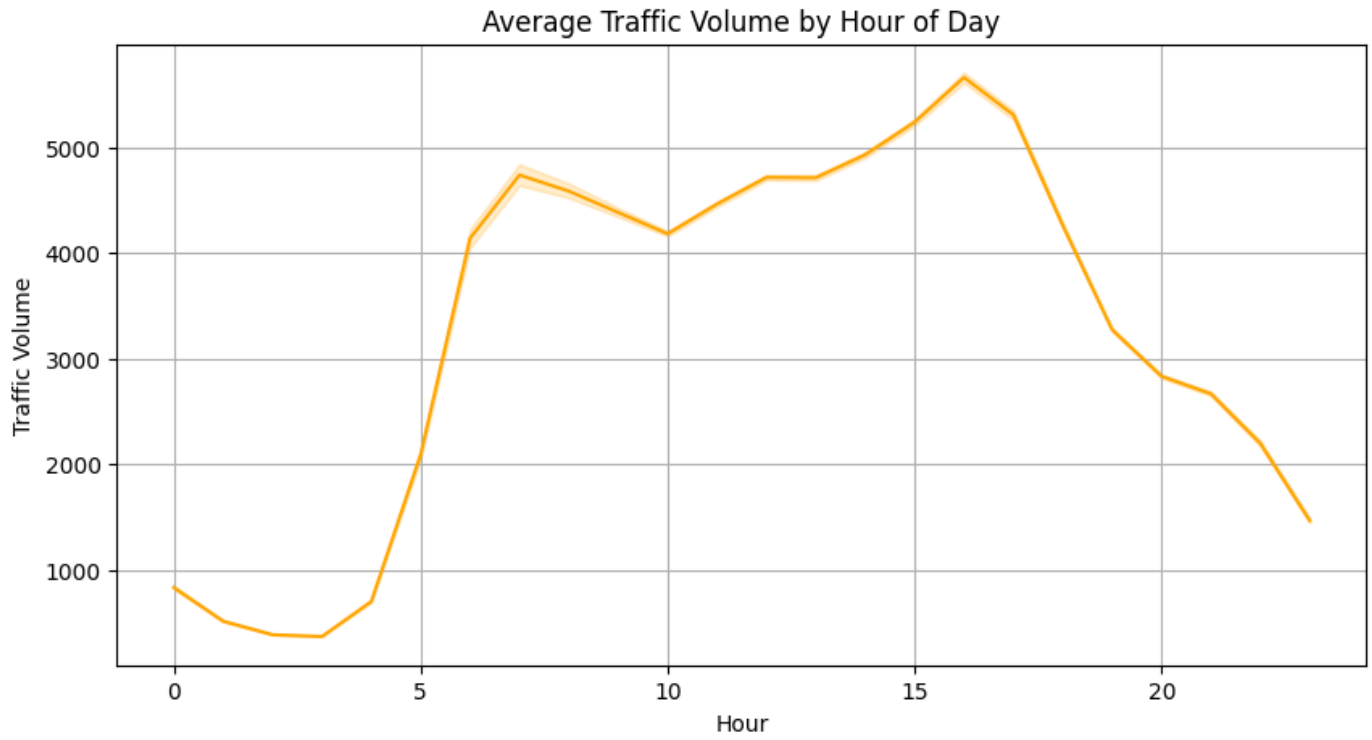
Findings of this Project:

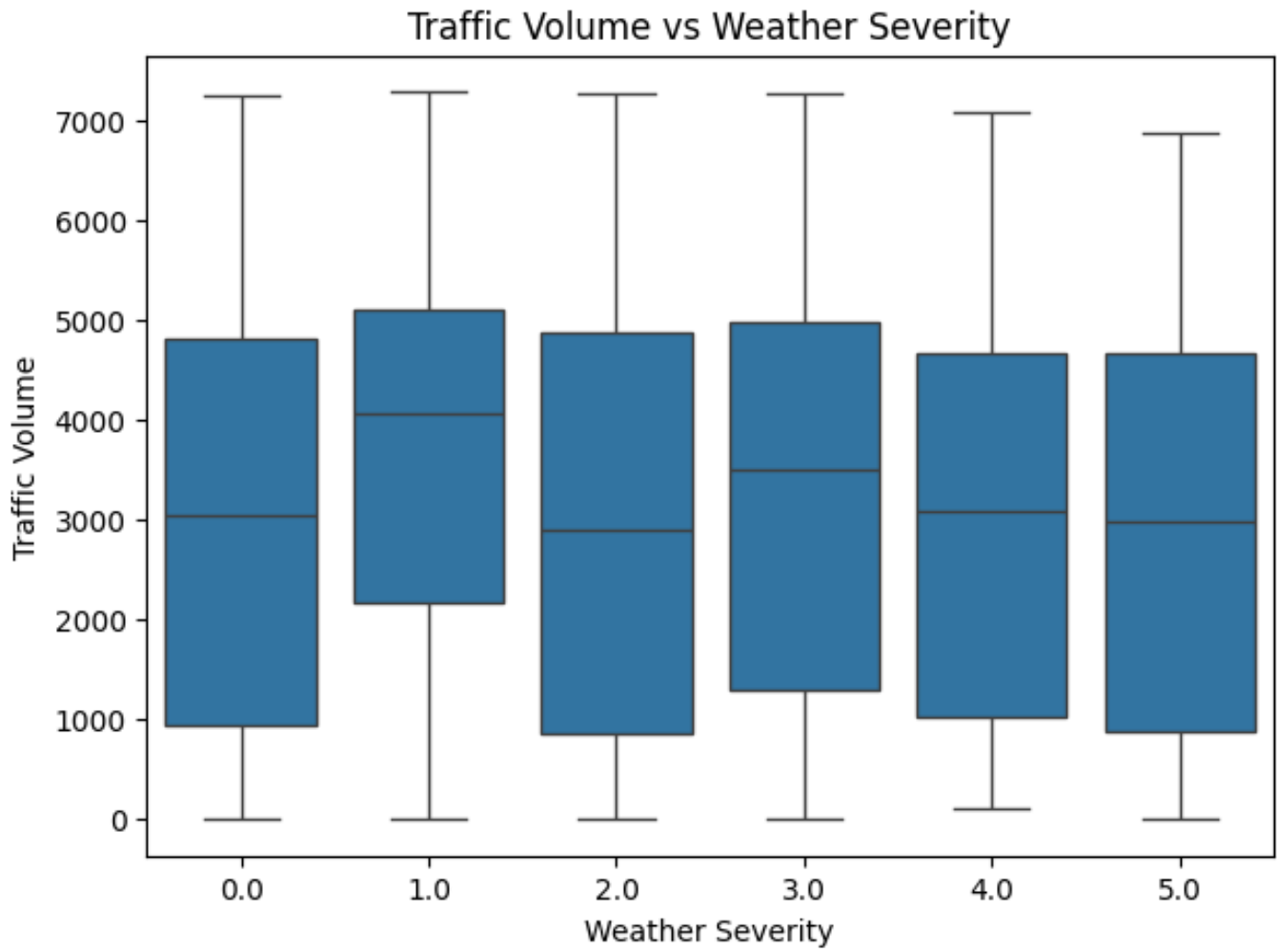
- PCA provides an unbiased and efficient way to create a risk score.
- Random Forest outperforms other models, providing stable and high accuracy.
- KMeans clustering reveals clear condition patterns that act as hotspots.
- A simple React–Flask interface can make real-time risk prediction practical.

Traffic Volume Distribution





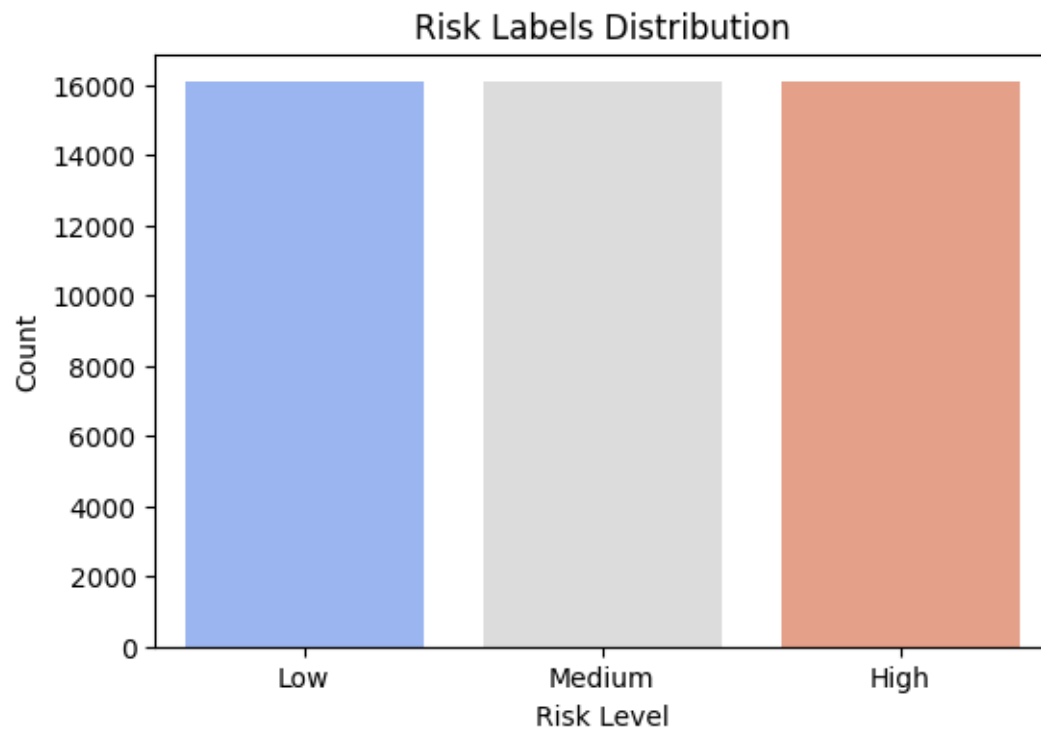


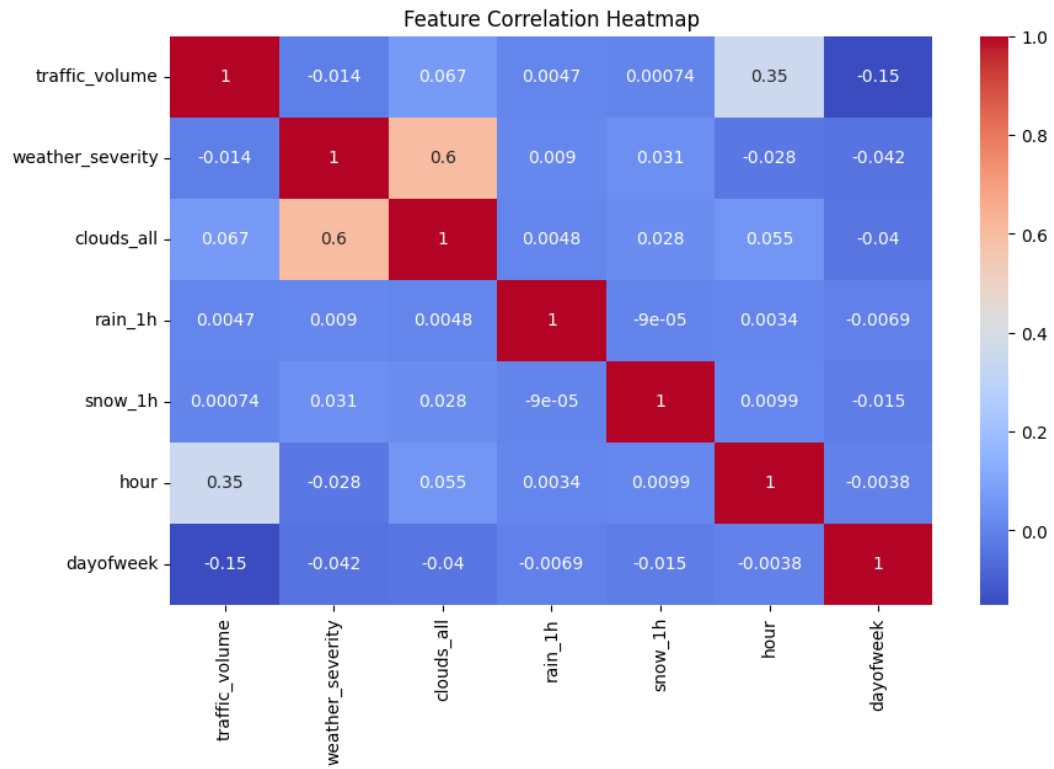


4. Methodology

- The project follows a structured machine-learning pipeline:
- **Data Collection:**
Metro Interstate Traffic Volume dataset (traffic, weather, time features).
- **Preprocessing:**
 - Removal of duplicates
 - Missing value handling
 - Weather severity mapping
 - Time extraction (hour, day of week)
 - Feature scaling with StandardScaler
- **PCA Risk Scoring:**
PCA applied on traffic and weather features → generated a single “risk score.”
The score was divided into Low/Medium/High categories using quantiles.
- **Supervised Model Training:**
 - Logistic Regression
 - Random Forest

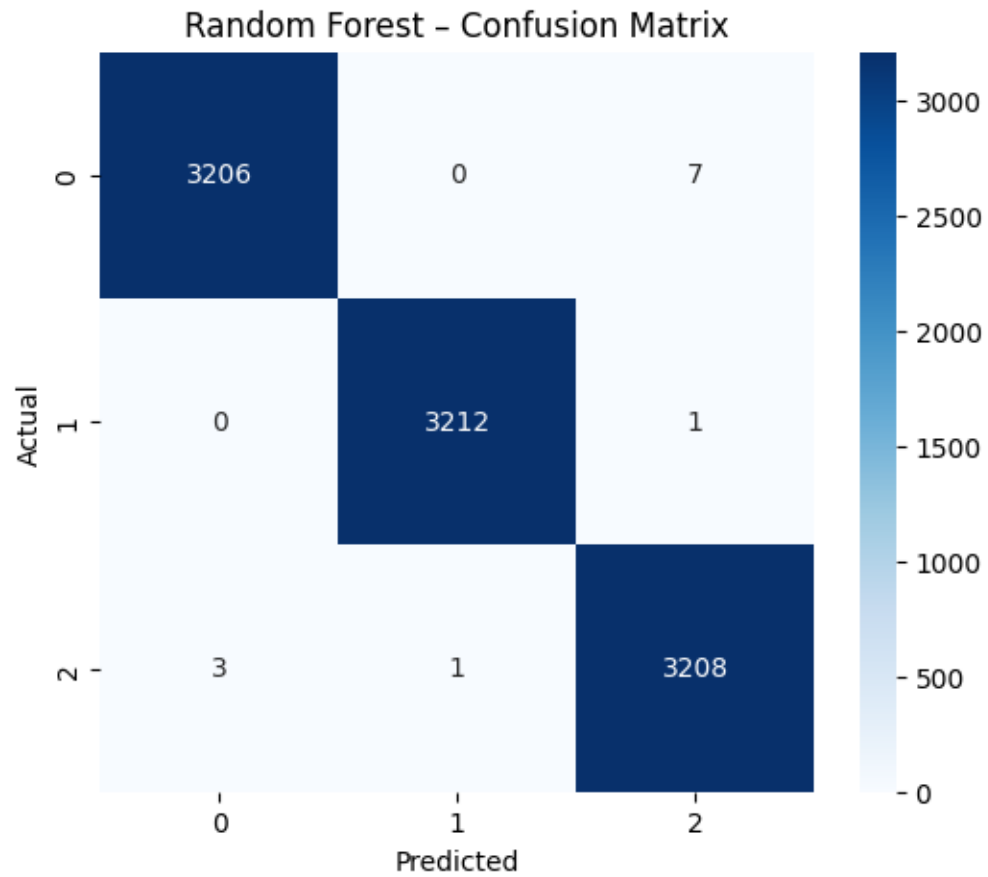
- SVM (RBF kernel)
Train-test split (80–20) and model evaluation using accuracy and classification report.
- **Hotspot Clustering:**
KMeans (k=3) used to group similar conditions into natural clusters.
- **System Development:**
 - Frontend: React-based interface
 - Backend: Flask REST API
 - Model deployment using joblib





5. Results and Discussion

- • PCA successfully captured the dominant pattern linking traffic and weather to risk.
- ☐ **Random Forest achieved the highest accuracy**, outperforming Logistic Regression and SVM.
- ☐ The classification report showed well-balanced precision and recall across all three risk levels.
- ☐ KMeans clustering produced three clear hotspot groups, revealing patterns such as high traffic + severe weather forming high-risk hotspots.
- ☐ The React–Flask application allowed users to instantly predict current risk and see the hotspot category of the entered conditions.
- ☐ The system demonstrates practical use for early warning and situational awareness.



MODEL ACCURACY COMPARISON:

	Model	Accuracy
0	Logistic Regression	0.988587
1	Random Forest	0.998755
2	SVM	0.990973

6. Conclusion

This project successfully developed a complete machine-learning system for accident risk prediction and hotspot analysis.

PCA removed manual weighting bias and helped generate meaningful risk labels.

Random Forest proved to be the most reliable model for classification.

Clustering enhanced interpretability by identifying risk patterns in the dataset.

The full-stack interface allowed real-time user interaction, making the system practical and accessible.

7. Future Work

- Integrate **real-time weather and traffic APIs** for live prediction.
- Add **map-based hotspot visualization** using Google Maps or Leaflet.
- Deploy system on **AWS/Azure/GCP** for public access.
- Explore advanced models (Gradient Boosting, Deep Learning).
- Build a mobile-friendly UI for on-road users.
- Incorporate alert/notification systems for high-risk conditions.